

Hands on 5 - Clean Up & Push to Remote Git

Step 1: Create a New Repository

```
mkdir GitRemoteDemo  
cd GitRemoteDemo  
git init
```

Explanation:
Creates a new Git repository.

Step 2: Create the Initial File

Commands:
echo "Welcome to Remote Git" > welcome.txt
cat welcome.txt

Explanation:
Creates welcome.txt and verifies its contents.

Step 3: Stage and Commit

Commands:
git add welcome.txt
git commit -m "Initial commit"

Explanation:
Creates the first commit.

Step 4: Verify Clean State

Commands:
git status
git branch

Explanation:
Ensures the repository is clean and lists the current branch.

Step 5: Connect to GitHub

Commands:
git remote add origin https://github.com/<your-username>/GitRemoteDemo.git
git remote -v

Explanation:
Connects the local repository to the GitHub repository.

Step 6: Rename Default Branch

Command:

```
git branch -M main
```

Explanation:

Renames the default branch to main.

Step 7: Push to GitHub

Command:

```
git push -u origin main
```

Explanation:

Pushes the local repository to GitHub and sets the upstream branch.

If prompted, enter your GitHub username and use a Personal Access Token (PAT) as the password.

Step 8: Pull Latest Changes

Command:

```
git pull origin main
```

Explanation:

Downloads any new commits from GitHub.

Step 9: Verify

Commands:

```
git status
```

```
git log --oneline --graph --decorate
```

Explanation:

Verifies that the repository is clean and displays the commit history.

Output:

```
GitRemoteDemo — -zsh — 80x21

[Password for 'https://adityahargunani@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 245 bytes | 245.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/adityahargunani/GitRemoteDemo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
adityahargunani@Adityas-MacBook GitRemoteDemo % git pull origin main
From https://github.com/adityahargunani/GitRemoteDemo
 * branch            main       -> FETCH_HEAD
Already up to date.
adityahargunani@Adityas-MacBook GitRemoteDemo % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
adityahargunani@Adityas-MacBook GitRemoteDemo % git log --oneline --graph --deco]
rate
 * df3f9aa (HEAD -> main, origin/main) Initial commit
adityahargunani@Adityas-MacBook GitRemoteDemo % █
```

